

COMPUTER SCIENCE TRIPOS Part IB – 2024 – Paper 5

6 Introduction to Computer Architecture (swm11)

ECAD+Arch
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The SystemVerilog code below is for a simple input/output processor that is focused on handling state changes on input and output pins.

```
typedef enum { CONST, JMP, BNZ, PUSH,
               SET, GET, INC, DEC } opcode;
typedef enum { RX, RY } regname;
typedef logic [31:0] word;
typedef logic [3:0] immediate;
typedef logic [3:0] pins;
typedef struct {opcode op;
               regname r;
               immediate i; } instruction;

module pio_proc
(
    input logic clk,
    input logic rst,
    input pins inpins,
    output pins outpins,
    output word outw
);
    instruction imem[0:15] =
        '{op:CONST, r:RY, i:4'd0}, // 0
        '{op:GET, r:RX, i:4'd3}, // 1
        '{op:DEC, r:RX, i:4'd0}, // 2
        '{op:BNZ, r:RX, i:4'd1}, // 3
        '{op:GET, r:RX, i:4'd3}, // 4
        '{op:BNZ, r:RX, i:4'd4}, // 5
        '{op:INC, r:RY, i:4'd0}, // 6
        '{op:GET, r:RX, i:4'd3}, // 7
        '{op:DEC, r:RX, i:4'd0}, // 8
        '{op:BNZ, r:RX, i:4'd6}, // 9
        '{op:INC, r:RY, i:4'd0}, // 10
        '{op:GET, r:RX, i:4'd3}, // 11
        '{op:GET, r:RX, i:4'd3}, // 12
        '{op:BNZ, r:RX, i:4'd10}, // 13
        '{op:PUSH, r:RY, i:4'd0}, // 14
        '{op:JMP, r:RX, i:4'd0}}; // 15

    instruction ir;
    immediate pc;
    word rf[regname];
    always_comb ir = imem[pc];
    always_ff @(posedge clk or posedge rst)
        if (rst)
            pc <= 0;
        else begin
            case(ir.op)
                CONST: rf[ir.r] <= {28'b0, ir.i};
                JMP: pc <= ir.i;
                BNZ: pc <= (rf[ir.r] != 0) ? ir.i : pc+1;
                PUSH: outw <= rf[ir.r];
                SET: outpins[ir.i[1:0]] <= rf[ir.r][0];
                GET: rf[ir.r] <= {31'b0, inpins[ir.i[1:0]]};
                INC: rf[ir.r] <= rf[ir.r]+1;
                DEC: rf[ir.r] <= rf[ir.r]-1;
            endcase // case (ir.op)
            if ((ir.op != JMP) && (ir.op != BNZ))
                pc <= pc+4'd1;
        end
    endmodule // pio_proc
```

- (a) How many clock cycles does it take to fetch, decode and execute an instruction? Explain your answer. [2 marks]

Answer: Every instruction completes in exactly one clock cycle including all steps: fetch, decode and execute.

- (b) What are the eight instructions and what function do they each perform? Give a short English explanation of each. [6 marks]

Answer: Note: assignment of marks are indicated in []

- JMP — jump to the immediate [0.5]
- BNZ — if register `ir.r` is not zero, jump to the immediate [1]
- CONST — register `ir.r` is assigned the zero extended immediate `ir.i` [1]
- PUSH — `outw` is assigned the register `ir.r` [0.5]
- SET — set pin specified by the lower two bits of the immediate (`ir.i[1:0]`) to the LSB of register `rf[ir.r]` [1]
- GET — get the pin specified by the lower two bits of the immediate (`ir.i[1:0]`), zero extend and write to register `rf[ir.r]` [1]

- INC — register `ir.r` is incremented [0.5]
 - DEC — register `ir.r` is decremented [0.5]
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- (c) The `imem` memory is initialised with a program. Produce commented pseudo code that represents the program. [6 marks]
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Answer:

```
while(True) {
    RY = 0
    do {
        RX = inpins[3] - 1;
    } while(RX!=0); // wait for inpins[3]==1
    do {
        RX = inpins[3];
    } while(RX!=0); // wait for inpins[3]==0
    do {
        RY++
        RX = inpins[3] - 1;
    } while(RX!=0); // wait for inpins[3]==1
    do {
        RY++
        RX = inpins[3];
    } while(RX!=0); // wait for inpins[3]==0
    outw = RY          // output result
}
```

- (d) In the SystemVerilog module, what registers are not reset and will they cause the processor to malfunction? [3 marks]
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Answer: Registers `outpins`, `outw` and `rf` are not reset. For the provided program, the `rf` registers do not need to be reset because they are written before being read. The `outw` data will be undefined until the first value is written to it, which is not ideal but will not cause the program to malfunction. `outpins` are not used by this program. Note that `ir` and `inpins` are wires not registers.

- (e) What is the meaning of the `outw` output? Is the instruction timing of the program important? [3 marks]
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Answer: `outw` represents the frequency of the input pin 3. The instruction timing is important: four instructions in each of the two loops that count loops between input edges. Thus, if the `pio_proc` clk frequency is f then $\text{outw} = f/4$
